

Asmita A

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EDUCATION

Manipal Institute of Technology, Manipal

2024 – 2028

Bachelor of Technology – Computer Science & AI (Machine Learning)

CGPA: 9.24

- Certificate: Algorithmic Toolbox – University of California San Diego (Coursera, Jan 2025)

EXPERIENCE

Web Development Intern

June 2025 – July 2025

Avaniko Technologies

Chennai, India

- Built RESTful APIs using ASP.NET Core and C#.
- Contributed to frontend UI components using HTML, CSS, and React.js.

AI & BI Intern

Month Year – Month Year

Winspire Solutions

Location

- Built Power BI financial report templates using ETL processes, DAX functions, and Microsoft Business Central data sources.
- Trained and tested AI agents in Microsoft Copilot Studio, including a Sales Proposal agent.

Data Structures Research

August 2025 – Present

Under Faculty Guidance, MIT Manipal

Manipal, India

- Ongoing research on optimized word lookup data structures for search strategies.

PROJECTS

Syncify – Full-Stack Scheduling App | *React, Node.js, Express, MySQL, JWT*

GitHub | Live Demo

- Built a multi-user calendar and scheduling web app with a chatbot assistant that finds common free time slots across participants.
- Automated email reminders using node-cron and implemented JWT-based authentication with REST APIs.
- Implemented MySQL database functionalities including triggers and stored procedures to support scheduling logic.

LPS Service Center Backend – Supreme Computers | *Python, Django*

- Developed a backend system using Django to manage laptop inventory and servicing for Supreme Computers, Chennai, promoting migration from outdated manual/legacy software.
- Improved registration and challan generation time, streamlining day-to-day service center operations.

WORDJAM – AI-Powered Semantic Crossword Game

Flask, JS, Groq | GitHub | Live Demo

- Built a full-stack crossword game with a custom grid-generation and word-fitting engine, AI-generated clues, and semantic warm/cold guess feedback.
- Implemented grid generation via recursive backtracking with flood-fill connectivity validation, and a word-fitting solver using MRV-heuristic backtracking; optimized lookups with a precomputed hash index, replacing O(n) scans over 41,000 words.
- Engineered semantic feedback using cosine similarity of sentence embeddings, and integrated the Groq LLM API for AI-generated clues with prompt-engineered guardrails.

Gesture-Controlled Car Racing Game | *Python, OpenCV, MediaPipe*

- Built a car racing game controlled entirely by hand gestures, using MediaPipe hand-tracking to map real-time hand position and tilt to steering and acceleration inputs.

MERN Shopping App | *MongoDB, Express.js, React, Node.js*

- Developed and deployed a full-stack e-commerce application with JWT-based user authentication, product catalog, and cart functionality.

TECHNICAL SKILLS

Languages: C++, Python, Java, C#, JavaScript

Frontend: React.js, HTML, CSS, Tailwind CSS

Backend: Node.js, Express.js, ASP.NET Core, Django

Databases: MongoDB, SQL

Languages Spoken: English, Tamil, Hindi

ACHIEVEMENTS

CGPA: 9.24 | Branch Rank 2/237 (Semester 1) – MIT Manipal